



Capstone Faculty Mentor Briefing Report Template

Report last updated: January 15th, 2026 (01/15/2026)

Instructions:

The purpose of this document is to summarize progress made over the previous term so the faculty mentor can get caught up quickly. Activities last term should have included at least 1-2 meetings with the sponsor to initiate relevant onboarding and training, discovery, obtaining access to data, and a brief data review. Things may change over the course of capstone, and this information can evolve. The information can also be carried forward to the Progress Reports submitted later.

This document is to be emailed to the faculty mentor in the first week of the capstone course. It will be graded according to the rubric below **in green font**.

TOTAL POINTS: 10

I. PROJECT BACKGROUND

[B1] (1 POINT) Stakeholder Names and Roles

In the table below, enter information for each team member, mentor, and sponsor. On the sponsor side, there may be several stakeholders including project manager, data contact, etc.

Stakeholder	Role
Ryan Healy	Team member
Louis Cocks	Team member
Katherine Kelleher	Team member
Anson Parker	Sponsor Role
Lane Rasberry	Sponsor Role
Rafael Alvarado	Capstone Faculty Mentor

[B2] Project Title: Process Modeling in Wikipedia Arbitration Disputes

[B3] (1 POINT) Abstract

The objective of this project is to develop an automated framework for extracting formal ontologies and generating clear executable process models from Wikipedia Arbitration Disputes. The research addresses the challenge of translating the raw Wikipedia disputes into machine-readable representations suitable for compliance checking, workflow automation, and insight into the differences between stated and actual processes.

II. DISCOVERY

[D1] (3 POINTS; full credit given if attempt was made AND details are provided below)

Did some subset of the student team meet with the sponsor? If so, briefly summarize the discussion. Include at a minimum to the best of your ability:

- The business purpose of the project

This project will demonstrate the use of Ontology Extraction and Business Process Modeling & Notation (BPMN) within a scoped and well-understood domain. The objective is to extract processes for Wikipedia arbitration by scoping documents as well as extracting and mining data, and use subject-matter-expertise to verify processes. For Wikipedia sources, if two individuals report differing information dispute information-

- How your work will be used in practice. By working backwards from the goal, you will be more likely to solve the right problem.

In practice, this project will help to provide Wikipedia users clarity on the way disputes are often handled and demonstrate a proof-of-concept analytical solution that can be generalized that can extract information

- Why the project is important

A proof-of concept solution and/or methodology that supports process extraction from extensive text data, many individuals, and various rules will provide value by demonstrating how a technical solution can collect and analyze information at scale to provide insight into processes that would otherwise be manually intensive or impossible to interpret.

- Who are the stakeholders

The stakeholders include Anson Parker (Lexipedia Project Lead) and Lane Rasberry (Wikipedia Dispute Subject-Matter-Expert). Anson Parker is looking to develop an analytical solution that

demonstrates process extraction. Lane Rasberry has knowledge about the Wikipedia dispute process.

- Anything that is unclear about the problem

As the team has not started to parse available Wikipedia data, the full scoping of the generalized solution/methodology that will be included in the final project deliverable has yet to be finalized.

- *Important assumptions*

Every arbitration, no matter how big or small, inherently has some level of BPMN behind its arbitration. Arbitration is not solely comprised of argumentative statements.

Wikipedia has arbitration data that is accessible via API during the timeline of the project.

- *What is in scope / out of scope (as needed)*

Any information within the Wikipedia Arbitration data is available and may be included in scope for process extraction (excluding non-English examples to help maintain consistency).

- *What are the success criteria*

- The team is able to show a technical, proof-of-concept solution or methodology that can extract an end-to-end arbitration process from Wikipedia
- The solution/methodology has an analytical capability that generates insights from all process information extracted from Wikipedia
- The solution has generalizable components that have the potential to facilitate the extraction and analysis of other processes from text data

[D2] (3 POINTS; full credit given if attempt was made AND details are provided below)

Did you receive data? If so, briefly describe the data. If not, explain what prevented receipt of the data.

No, due to a lengthy scoping process and a late scope change, the team did not receive a designated dataset. The data the team will now be working with is open and easily accessible Wikipedia data that Lane will help the team to properly understand and use. The data is free to scrape or pull from the Wikipedia API and in the coming weeks Lane will help the team to better understand the nuances.

[D3] If you received the data, did you have a chance to review it? Did you notice any potential challenges with the data?

The Wikipedia arbitration data is well structured, but the processes that generate it might have changed over the last 20 years. The data within these sources can vary greatly from using every arbitration process at disposal (like arbitration on “Ethics or War”) to two users disagreeing over and adjudicating the correct date that is relevant to an event in history (may use one arbitration process).

III. COMPUTING

[C1] (1 POINT) Given what you know about the data, what do you plan to use for data storage (e.g., Google Drive, Rivanna)?

The team can store the data locally and in Google Drive to start. As we scale, we can move our data to the Rivanna/Afton cluster as needed.

[C2] (1 POINT) Given what you know about the data, what do you plan to use for computing (e.g., Google Colab, your local machine, Rivanna)?

The team will begin with local computers and in-memory analysis to start and will request a Rivanna/Afton allocation as needed for scaling.

IV. MISC

Anything else you would like to share with the faculty mentor?

The most recent sponsor engagements have involved a huge scope change from original project discussion that is still being fully defined. The collaborative scoping process between stakeholders was necessary to ensure the final deliverable will provide stakeholder value. The original scope involved Lexipedia directly (web scraping legal sources like VA state law sites, building ontologies from the data, and training models to build out that network). The current, different scope involves a set data source and access points (Wikipedia Arbitration Data) that should better serve to build out BPMN modeling.